AESH SUSTAINABILITY HACKATHON 2026

# CHALLENGE 1 FINALIST GUIDE

Sustainable Electronics for Space Systems

**FINAL PHASE REQUIREMENTS AND DEADLINES**

**Onsite Final - Egypt | 6-7 July 2026**

**COMMITTEE REVIEW DRAFT**

**Final Phase objective**

Develop one bounded low-power electronic subsystem for a realistic space mission and prove the net power or energy improvement.

# 1. What Your Team Must Build

- One clearly defined subsystem - not a complete satellite.
- A system architecture showing inputs, outputs, power/data flow, and operating modes.
- A fair baseline power/energy case with units and traceable assumptions or measurements.
- An implemented optimized mode such as sleep, duty cycling, adaptive sensing, or power control.
- Net savings calculation including controller, sensor, communication, and decision-logic overhead.
- Testing across multiple operating cases plus one clear limitation/trade-off.
- A live hardware demo or executable model that shows the core operating-mode change.

# 2. Required Final Deliverables

| Deliverable | Requirement |
|---|---|
| Updated Project Repository | Accessible to judges; organized files; README with setup, execution, parameters, assumptions, and reproduction steps. |
| Working Proof of Concept | Hardware, software, executable simulation, or validated analytical model demonstrating the core function. |
| Final Evidence Sheet | One page: problem, core function, baseline, improved case, test conditions, primary technical KPI, primary sustainability KPI, limitation, and repository link. |
| Final Presentation | No fixed slide count. Use the number of slides that fits a 5-minute presentation and keeps text readable. |
| Live Demo | Maximum 5 minutes; must show the core function and main evidence. |
| Backup Demo Video | 60-90 seconds; used only if a genuine technical failure occurs. |
| Equipment and Safety Declaration | All power, hardware, RF, acoustic, mechanical, battery, and special-space needs must be declared and approved. |
| AI / External Resource Disclosure | List meaningful AI assistance, libraries, datasets, prior work, datasheets, standards, and external technical sources. |

# 3. Challenge-Specific Repository Evidence

- Source code, circuit/schematic, notebook/model, or measurement files.
- Component and parameter table with sources/datasheets.
- Baseline and optimized power/energy table.
- Plots or logs from validation tests.

# 4. Recommended Primary KPIs

- Average/peak current or power
- Energy per operating cycle
- Duty cycle or active-time reduction
- Estimated effect on battery, solar, thermal, or mission resources

# 5. Official Deadlines

**Time zone**

All deadlines are in Egypt Summer Time (EEST, UTC+3). A written update from the organizing committee overrides this table only if issued officially.

| Deadline | Required action |
|---|---|
| 17 June 2026 - 23:59 EEST | Finalist confirmation, onsite attendance confirmation, and final team roster. |
| 20 June 2026 - 23:59 EEST | Repository-access confirmation plus Equipment and Safety Declaration. |
| 23 June 2026 - 23:59 EEST | One-page Final Phase Implementation Plan: scope, core function, KPI, test plan, and main risk. |
| 30 June 2026 - 23:59 EEST | Progress proof: 60-90 second progress video, one technical result, and one sustainability result. |
| 3 July 2026 - 23:59 EEST | Final digital package: updated repository, evidence sheet, presentation, backup demo, and disclosures. |
| 6 July 2026 | Onsite Day 1: opening, workshops, preparation, networking, and related activities. Detailed agenda to be announced later. No official judging. |
| 7 July 2026 - 12:00 EEST | Final repository tag/file freeze and final presentation confirmation. |
| 7 July 2026 - 13:30 EEST | Official Final Phase judging begins. |

## 6. Final-Day Format

- Day 1 (6 July): opening, workshops, preparation, networking, and related activities. Detailed agenda will be announced later. No official judging.
- Day 2 judging begins at 13:30.
- Each team receives 15 minutes: 5-minute presentation, 5-minute live demo, 4-minute Q&A, and 1-minute scoring/transition.
- The panel includes the designated Technical Judge for your challenge and two Sustainability Reviewers who evaluate every challenge.
- Final ranking uses 75% technical and 25% average sustainability. Only First, Second, and Third Place are awarded.

## 7. What Is Not Sufficient

- Repeating the Phase 1 concept without new implementation or validation.
- A promotional video with no technical evidence.
- An unexplained percentage improvement or a baseline that performs a different task.
- Screenshots without runnable files, parameter values, units, or interpretation.
- A complex project that cannot demonstrate one stable core function.
- Using AI-generated or external work that the team cannot explain, verify, or disclose.

## 8. Rules and Safety

- Continue the same challenge and core project qualified in Phase 1. Scope may be refined, but the project may not be replaced.
- Only registered team members may present, operate the demo, and answer questions.
- Unauthorized RF transmission, high-power acoustic transmission, hazardous batteries, lasers, chemicals, pressure systems, or unsafe mechanical motion are prohibited.
- Teams retain responsibility for originality, licensing, data accuracy, and full understanding of their submission.
- The organizing committee may stop unsafe or non-compliant demonstrations.